Let us add, remove, and edit a file from the repository, and verify the status:

```
$ echo "This is sample file" >  sample-file.txt
$ echo "Jarvis" >> CONTRIBUTING
$ rm COPYING
$ git status -s
```

Now the repository status has been changed. The status command will show the following output:

```
M CONTRIBUTING
 D COPYING
?? sample-file.txt
```

In the above output, the letter 'M' in the first column indicates that there has been a modification to the 'CONTRIBUTING' file. The letter 'D' indicates that the 'COPYING' file has been deleted. The '??' indicates that *sample-file.txt* is the new file and Git doesn't know anything about it.

To verify the changes made in the repository, execute the *git diff* command as follows:

```
$ git diff
```

This command will show output in the UNIX pager. The lines in green with the '+' symbol at the beginning represent the new addition to the repository. The lines in red with the '-' symbol at the beginning represent deletion from the repository. Figure 3 is the screenshot of the same.

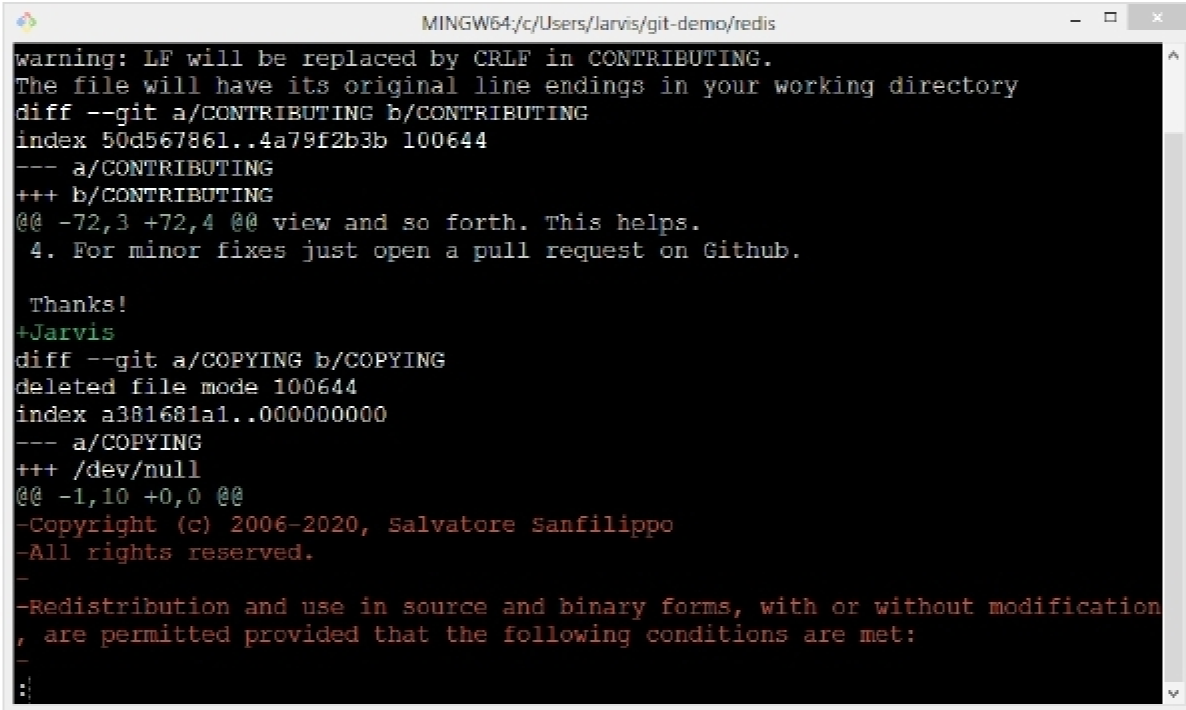


Figure 3: Output of git-diff command

## Upload files to the repository

In the previous section, we made a few changes to the repository. However, those changes are still on the local file system. Let us upload them to the remote repository.

First, create a changeset. We will include all the three files in the changeset as follows:

```
$ git add CONTRIBUTING COPYING sample-file.txt
$ git status -s
```

The above command will generate the following output:

```
M  CONTRIBUTING
D  COPYING
A  sample-file.txt
```

If you see carefully, there is the letter 'A' before the *sample-file.txt* file, which indicates that it's a new file in the repository and part of the current changeset. Now, let us create a commit:

```
$ git commit -m "Sample commit"
```

This command will generate the following output:

```
[unstable be159ed3d] Sample commit
 3 files changed, 2 insertions(+), 10 deletions(-)
 delete mode 100644 COPYING
 create mode 100644 sample-file.txt
```

Let us verify the commit using the *log* command as follows:

```
$ git log --oneline
be159ed3d (HEAD -> unstable) Sample commit
```

Wonderful! Our commit appears in the repository. Also, a unique SHA-1 hash is generated and assigned to it.

However, our changes are still present on the local file system. To send this commit to the remote repository, execute a *push* command as follows:

```
$ git push
```

**Note:** Connectivity to the remote Git server is needed to perform this operation.

## Syncing the repository

In the previous section, we uploaded our changes to the remote repository. Similarly, other developers will upload their changes to the remote repository too. Hence, we should sync our local repository periodically. We can use the *pull* command for this:

```
$ git pull
```

**Note:** Connectivity to the remote Git server is needed to perform this operation too.

END

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